

User Manual

MatrixClock Improved Firmware v3.1.2



1. Quick start first installation

Requirements

Original HACK LABS MatrixClock-compatible ESP8266 hardware with 4 MB flash, a regulated 5 V USB supply, and a 2.4 GHz Wi-Fi network.

1. Power the clock. A clean installation starts the open **MatrixClock** Wi-Fi network.
2. Join **MatrixClock** on a phone or computer and open **<http://192.168.4.1/>**.
3. Select the home Wi-Fi network, enter its password, and choose **Save and reboot**.
4. Read the local IP address that scrolls during boot, then open that address in a browser.
5. Choose a web username and password, or leave both blank to disable local web authentication.

Boot display

Normal startup shows **LOAD - WiFi - OK! - NTP - OK!**, then the local IP address and clock. If NTP fails, **Err!** is shown; RTC time continues when valid, and **No sync-Set NTP** is displayed twice after five seconds.

2. Daily operation button and web controls

Physical button

Control	Behaviour
Clock: short press	Scroll the configured date marquee.
Clock: hold about 1 second	Enter Chronograph mode.
Chronograph: short press	Start, stop, or resume timing.
Chronograph stopped: hold	Reset to zero; when already at zero, exit to the clock.
Hold about 10 seconds	Start the protected RESET warning and factory-reset sequence.
While an API message scrolls	One press finishes the current pass gracefully; a second press within one second returns directly to the clock.

Web interface

Area	What it provides
Main page	Live 32 by 8 display preview, Device info, clock-accuracy calibration, API message test and message cancellation.
Settings	Display, timezone, NTP provider, reboot mode, security, drift reset, Factory reset and Firmware update.
Display settings	Brightness, speed, date interval, separator, transition, Chronograph format, live-view colour and message scroll count. Save together without reboot.
Time settings	Timezone and NTP changes apply after Save and reboot. Forty regional presets plus automatic, fixed-standard-time and custom POSIX modes are available.
Security	Optional local username/password protection. API callers use the same HTTP Basic credentials. The setup AP remains open.

Useful details

The tilt switch keeps the physical display upright. The web preview supports two simultaneous main-page viewers. Device info includes uptime, NTP details, calibration state, Chronograph history and DS3231 RTC temperature.

3. Accuracy and integration NTP, RTC and Home Assistant

How timekeeping works

Function	Behaviour
NTP checks	The clock synchronises at startup, settles, then performs regular checks using filtered low-delay replies.
Smooth alignment	Accepted phase errors are corrected gradually so the displayed time does not visibly jump.
Drift calibration	Successive stable measurements estimate oscillator error. A qualified correction is applied automatically and saved safely.
RTC fallback	The DS3231 retains local time and date if NTP is unavailable. A saved drift correction is used when available, and NTP retries continue.
Chronograph	Uses rollover-safe elapsed timing and preserves its running correction until the session ends.

Calibration card

Correction in use is the active oscillator-rate adjustment. **Last NTP offset** shows how far the display was ahead of or behind NTP. **Measurement progress** shows progress toward stable evidence; learning and application are automatic.

Local HTTP API

```
POST http://CLOCK-IP/api/message
Content-Type: application/x-www-form-urlencoded
message=Bin day tomorrow&scrolls=2
```

Endpoint	Purpose
POST /api/message	Send up to 64 printable characters. Optional scrolls=1 to 5 overrides the saved count for that message.
POST /api/cancel	Finish the current pass and return gracefully to the clock.
GET /api/temperature	Return DS3231 internal temperature as JSON at native 0.25 C resolution.

Home Assistant

Reserve a fixed DHCP address, use a REST command for messages and a REST sensor for temperature. If security is enabled, store the MatrixClock credentials in **secrets.yaml**. Complete YAML examples are in the project README.

4. Firmware installation one binary for every supported route

Firmware file

Use **MatrixClock_Improved_v3_1_2_OTA.bin** for web updates and clean USB installations. Verify it against **SHA256SUMS.txt**. Keep power connected while writing firmware.

Update MatrixClock Improved

1. Open <http://CLOCK-IP/update> and sign in if security is enabled.
2. Select the v3.1.2 OTA file, choose **Upload and reboot**, and wait for restart.

Update directly from original HACK LABS firmware

1. Open <http://CLOCK-IP/update> and use username **nick**, password **nick**.
2. Upload the v3.1.2 OTA file and confirm MatrixClock Improved starts.
3. Open **Settings** and select **Factory reset**. This is recommended so obsolete original-firmware settings cannot remain in use. Then complete Wi-Fi setup again.

Clean USB installation

- Connect by USB and close any program using the clock's COM port.

```
esptool --chip esp8266 --port COM3 --baud 115200 erase_flash  
esptool --chip esp8266 --port COM3 --baud 115200 write_flash 0x000000 MatrixClock_Improved_v3_1_2_OTA.bin
```

- Replace COM3 with the correct port. After flashing, join MatrixClock Wi-Fi and open <http://192.168.4.1/>. No separate Factory image is required.

Quick troubleshooting

Symptom	Check
No web page	Use the IP shown at boot and ensure the browser is on the same local network.
Back in setup mode	Join MatrixClock Wi-Fi, open 192.168.4.1, and enter valid 2.4 GHz Wi-Fi details.
NTP Err!	RTC time can continue. Check internet access and the selected NTP provider in Settings.
Unstable display	Check the USB cable and use a known-good regulated 5 V supply.

Project and licence

Source, releases and full Home Assistant examples: github.com/maddenste/MatrixClock-Improved

Original project: github.com/hack-apollo/hack_clock

Modified and maintained by Steve Madden. Released under GNU GPL v3 without warranty.